

De novo Riboflavin Biosynthesis and Bacterial Flavin Cofactor Activation

A review-style synthesis for a molecular biology audience

1. Executive Summary

Riboflavin (vitamin B2) is the metabolic precursor of the flavin cofactors FMN and FAD, which serve as redox prosthetic groups for a large fraction of every organism's oxidoreductases. Bacteria that cannot import riboflavin from their environment must synthesize it *de novo*, and the pathway that does so is one of the most elegantly economical routes in primary metabolism: a single purine (GTP) and a single pentose-phosphate (ribulose 5-phosphate) are converted, through eight catalytic activities distributed over five or six gene products, into one molecule of riboflavin. The pathway is strictly ordered, has two converging feeder branches, and terminates in a remarkable macromolecular machine in which the last two enzymes are physically nested one inside the other.

The best-supported model is as follows. On the **pyrimidine branch**, GTP cyclohydrolase II (RibA) commits GTP to the pathway by opening the imidazole ring, releasing formate and pyrophosphate with the help of tandem active-site zinc and magnesium ions. The bifunctional deaminase/reductase RibD then deaminates and reduces the ribosylamino side chain in a fixed order (deamination *then* reduction in bacteria). On the **carbohydrate branch**, the magnesium-dependent 3,4-dihydroxy-2-butanone 4-phosphate (DHBP) synthase (RibB) excises C4 of ribulose 5-phosphate as formate to make the four-carbon xylene-ring donor. The two branches converge at **lumazine synthase (RibH)**, which condenses the pyrimidine with DHBP to make 6,7-dimethyl-8-ribityllumazine, and then at **riboflavin synthase (RibE)**, which dismutates two lumazines into one riboflavin, regenerating one pyrimidine substrate. In *Bacillus subtilis* the terminal step is carried out by a ~1 MDa complex — an icosahedral 60-subunit RibH capsid enclosing a RibE trimer. Finally, riboflavin is activated to the working cofactors by a single **bifunctional FAD synthetase (RibF/FADS)**, whose C-terminal module phosphorylates riboflavin to FMN and whose N-terminal module adenylylates FMN to FAD.

Three features make this system both scientifically interesting and practically important. First, its catalytic core is deeply conserved — bacteria and plants share reaction order, whereas fungi re-order the pyrimidine branch — making the bacterial enzymes a coherent, ancient module. Second, feedback control is exerted primarily at the RNA level by **FMN riboswitches (RFN elements)** rather than by classical protein repressors, with accessory protein inputs (e.g., the minor flavokinase RibR) layered on top. Third, because many pathogens lack an efficient riboflavin importer and the bacterial FMN-adenylyltransferase module is *non-homologous* to the human FAD-forming enzyme, the pathway offers a route to selective antibacterial agents — a rationale now validated by nanomolar substrate-analog inhibitors of lumazine and riboflavin synthases.

2. Definition and Biological Boundaries

What is included

This biological system comprises two tightly linked but conceptually distinct segments:

1. **De novo riboflavin ring biosynthesis** — the conversion of GTP and ribulose 5-phosphate into riboflavin through the ordered action of RibA (GTP cyclohydrolase II), RibD (deaminase + reductase), RibB (DHBP synthase), RibH (lumazine synthase), and RibE (riboflavin synthase).
2. **Flavin cofactor activation** — the conversion of riboflavin into FMN and then FAD by riboflavin kinase and FMN adenylyltransferase activities, which in most bacteria reside in the single bifunctional enzyme RibF/FADS.

The system's substrates are GTP (a purine nucleotide) and D-ribulose 5-phosphate (a pentose-phosphate-pathway intermediate); its products are FMN and FAD, the cofactors that feed the cellular flavoproteome.

What is often confused with it and should be treated separately

- **De novo purine biosynthesis.** The pathway *consumes* GTP but does not make it. Purine metabolism is a distinct, separately regulated pathway (with its own PurR repressor and guanine riboswitch); it is upstream and should not be folded into riboflavin biosynthesis, although engineering studies deliberately deregulate both together to boost flux ([PMID: 25023436](#)).

- **Cofactor F420 biosynthesis.** F420 is a deazaflavin made by a convergent but chemically different route; it should not be conflated with riboflavin biosynthesis ([PMID: 32570108](#)).
- **Triazine "secondary metabolite" branches.** Some bacteria divert GTP through RibA-like cyclohydrolases to make toxoflavin, fluviol, and related triazines. These share early enzymology with riboflavin biosynthesis but are antibiotic/toxin pathways, not cofactor supply ([PMID: 40044155](#)).
- **Riboflavin transport/uptake.** Import of exogenous riboflavin (via RibU/RibM/RfuABCD-type transporters) is an *alternative* route to the same cofactor endpoint and is regulated by the same FMN riboswitches, but it is mechanistically separate from biosynthesis ([PMID: 27822970](#)).
- **Downstream flavoprotein biology** (e.g., PutA, iron–riboflavin redox coupling) uses FAD/FMN but is not part of cofactor synthesis.

Competing definitions

The main source of ambiguity is **nomenclature**, not chemistry. Gene names differ between the two model organisms:

Activity	<i>E. coli</i> gene	<i>B. subtilis</i> gene
GTP cyclohydrolase II	<i>ribA</i>	<i>ribA</i> (part of bifunctional RibA/RibBA)
DHBP synthase	<i>ribB</i>	<i>ribB</i> (fused with cyclohydrolase in RibBA)
Deaminase/reductase	<i>ribD</i>	<i>ribG</i>
Lumazine synthase	<i>ribH</i> (<i>ribE</i> historically)	<i>ribH</i>
Riboflavin synthase	<i>ribC</i> (historically <i>ribB</i> / <i>ribE</i>)	<i>ribB</i> / <i>ribE</i>
Bifunctional FAD synthetase	<i>ribF</i>	<i>ribC</i>

The *B. subtilis* operon was originally described as *ribG–ribB–ribA–ribH–ribT* ([PMID: 8159171](#)), and the "RibA" of *B. subtilis* is bifunctional (RibA/RibBA), fusing GTP cyclohydrolase II and DHBP synthase. Readers must therefore map gene names to enzymatic activities carefully, because the same letter can denote different enzymes in different organisms.

3. Mechanistic Overview

The committed entry step

The pathway is committed by **GTP cyclohydrolase II (RibA)**. RibA performs a multi-step reaction on a single substrate: it hydrolytically cleaves the C8–N9 bond of the guanine ring, mounts a subsequent hydrolytic attack that liberates **formate**, and cleaves the α – β phosphodiester bond of the triphosphate to release **pyrophosphate**, yielding 2,5-diamino-6-(5-phospho-D-ribosylamino)pyrimidin-4(3H)-one. These steps are promoted by **tandem active-site metal ions** — **a zinc acting at the guanine and a magnesium at the triphosphate** (PMID: 34569786). Because this reaction irreversibly draws GTP into the pathway, it is the committed and canonically rate-influencing step.

The pyrimidine branch (RibD)

The RibA product is processed by the bifunctional **RibD** protein, which carries two activities on one polypeptide: **diaminohydroxyphosphoribosylaminopyrimidine deaminase** and **5-amino-6-(5-phosphoribosylamino)uracil reductase** (PMID: 23722836). In bacteria (and plants), **deamination precedes reduction** of the ribosyl side chain (PMID: 15208317). The product, 5-amino-6-ribitylamino-2,4(1H,3H)-pyrimidinedione, is the pyrimidine substrate for lumazine synthase.

The carbohydrate branch (RibB)

In parallel, **DHBP synthase (RibB, EC 4.1.99.12)** — a magnesium-dependent enzyme — **excises the C4 of D-ribulose 5-phosphate as formate**, producing (3S)-3,4-dihydroxy-2-butanone 4-phosphate (PMID: 35802469). DHBP is the four-carbon unit that becomes the xylene ring of the isoalloxazine system.

Convergence: lumazine synthase (RibH)

The two branches converge at **lumazine synthase (RibH)**, which condenses the pyrimidine with DHBP to form **6,7-dimethyl-8-ribityllumazine** (PMID: 7893702). This is the penultimate step of ring assembly.

The terminal dismutation: riboflavin synthase (RibE)

Riboflavin synthase (RibE) carries out a chemically unusual final step: it converts **two** molecules of 6,7-dimethyl-8-ribityllumazine into **one** molecule of riboflavin by an intermolecular **dismutation**, transferring a four-carbon unit from one lumazine to another and regenerating one molecule of the pyrimidine substrate (5-amino-6-ribitylamino-2,4(1H,3H)-pyrimidinedione), which is recycled ([PMID: 7893702](#)). This recycling reduces the net demand on the pyrimidine branch.

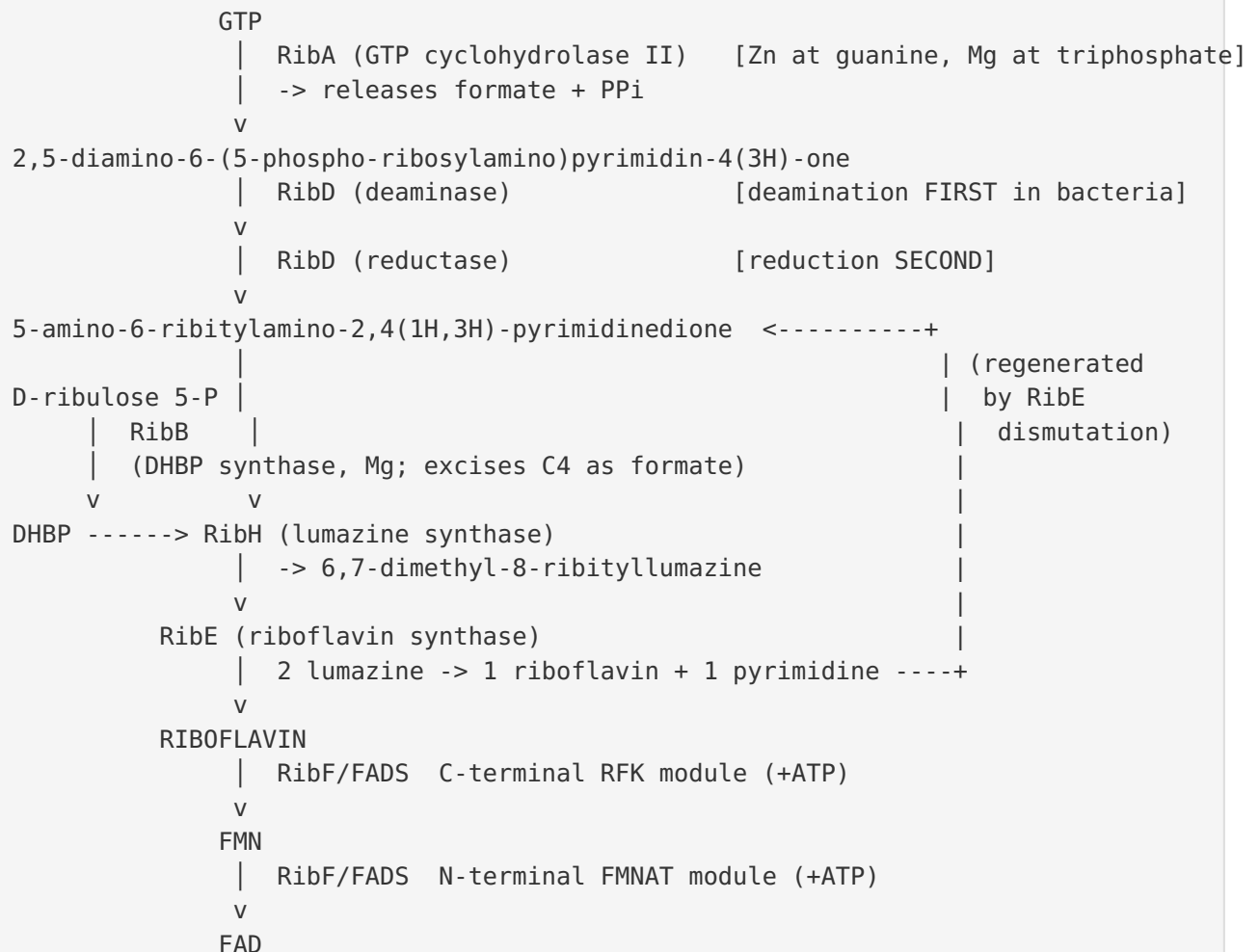
Cofactor activation: RibF/FADS

Riboflavin is then activated in two sequential ATP-dependent steps carried out, in most prokaryotes, by a **single bifunctional FAD synthetase (RibF/FADS)**:

1. The **C-terminal riboflavin kinase (RFK) module** phosphorylates riboflavin to FMN.
2. The **N-terminal FMN adenylyltransferase (FMNAT) module** adenylylates FMN to FAD.

A single enzyme, encoded by the *ribC/ribF* gene, thus carries out both activation steps ([PMID: 22892871](#)).

Sequence-of-events diagram



Obligatory vs. conditional vs. accessory steps

- **Obligatory and strictly ordered:** RibA → RibD(deaminase) → RibD(reductase) → RibH → RibE → RibF(RFK) → RibF(FMNAT). Each product is the next enzyme's substrate; the branch order within RibD (deaminate then reduce) is fixed in bacteria.
- **Conditional / parallel:** The RibB (carbohydrate) branch runs in parallel and must supply DHBP before RibH can act, but it is not "downstream" of the pyrimidine branch — the two are independent until they meet at RibH.
- **Accessory / regulatory:** FMN riboswitch control, the minor flavokinase RibR, and regulatory loci such as *ribU/ribC* modulate flux without being catalytic pathway members.

4. Major Molecular Players and Active Assemblies

The ~1 MDa lumazine synthase / riboflavin synthase complex

The most striking assembly in the pathway is the terminal *B. subtilis* complex, in which the last two enzymes are physically nested. It is a **~1 MDa bifunctional complex composed of 3 α (riboflavin synthase) subunits and 60 β (lumazine synthase) subunits; the 60 β subunits form an icosahedral capsid enclosing the α trimer in the central core** (PMID: 7473709). This architecture places the source of 6,7-dimethyl-8-ribityllumazine (RibH) around the enzyme that consumes it (RibE), providing a plausible substrate-channeling geometry.

The dismutation chemistry of the enclosed riboflavin synthase is itself notable: the lumazine intermediate is **"converted to riboflavin by the alpha subunits via an unusual dismutation yielding 5-amino-6-ribitylamino-2,4(1H,3H)-pyrimidinedione as second product"** (PMID: 7893702) — i.e., one lumazine donates a C4 unit to another, and the pyrimidine leaving group re-enters the pathway.

Metal-dependent branch enzymes

Two branch enzymes are metal-dependent in ways central to their chemistry:

- **RibA** uses **tandem Zn/Mg** to separate two chemically distinct tasks (guanine ring opening vs. triphosphate cleavage) within one active site (PMID: 34569786).
- **RibB** is a **Mg-dependent** enzyme that performs a skeletal rearrangement excising a carbon as formate (PMID: 35802469).

RibD: one protein, two active sites

RibD is a single bifunctional protein carrying both the deaminase and reductase activities; its structure is conserved across bacteria and it is described as a validated potential antibacterial drug target (PMID: 23722836).

The bifunctional FAD synthetase (RibF/FADS)

FADS is a two-module enzyme: **the RFK module is a homologue of eukaryotic monofunctional riboflavin kinases, while the FMNAT module lacks homology to eukaryotic FAD-producing enzymes** (PMID: 26627660). This split ancestry is the molecular basis for selective drug targeting: **the FMNAT step is carried out by two non-homologous proteins in eukaryotic and prokaryotic organisms** (PMID: 32466340). Structural and kinetic

work on the *Corynebacterium ammoniagenes* enzyme shows the two modules communicate through transient oligomerization (a "dimer of trimers"), with inter-protomer contacts modulating catalysis at both sites ([PMID: 28341845](#), [PMID: 25801930](#)).

Player-by-player summary table

Step	Enzyme (gene)	Activity	Key mechanistic feature	Assembly
Entry	RibA / RibBA (<i>ribA</i>)	GTP cyclohydrolase II	Tandem Zn/Mg; releases formate + PPi	Bifunctional with DHBP synthase in <i>B. subtilis</i>
Pyrimidine 1	RibD (<i>ribD/ribG</i>)	Deaminase	Deamination first (bacteria/plants)	Bifunctional, single polypeptide
Pyrimidine 2	RibD	Reductase	Side-chain reduction second	Bifunctional, single polypeptide
Carbohydrate	RibB / RibBA (<i>ribB</i>)	DHBP synthase	Mg-dependent; excises C4 as formate	Often fused to RibA in Firmicutes
Convergence	RibH (<i>ribH</i>)	Lumazine synthase	Condenses pyrimidine + DHBP	Pentamer → decamer → icosahedral 60-mer
Terminal	RibE (<i>ribE/ribC</i>)	Riboflavin synthase	Dismutation of 2 lumazines; recycles pyrimidine	Trimer (α_3), enclosed by RibH capsid
Activation 1	RibF/FADS (<i>ribF/ribC</i>)	Riboflavin kinase (RFK)	ATP → FMN	C-terminal module
Activation 2	RibF/FADS	FMN adenylyltransferase (FMNAT)	ATP → FAD	N-terminal module (non-homologous to human)

5. Evolutionary and Cell-Biological Variation

Conserved bacterial/plant core vs. divergent fungi

The catalytic core of the pyrimidine branch is deeply conserved between bacteria and plants: their **pyrimidine deaminase domains are structurally and functionally similar and use the same reaction sequence**, whereas **fungi use a different pathway** with the deamination/reduction order reversed (PMID: 15208317). This makes the bacterial reaction order (deaminate → reduce) a genuinely lineage-diagnostic feature, and it identifies the bacterial/plant enzymes as the best representatives of the ancestral pyrimidine-branch chemistry.

Gene architecture: fusion and fission

The same set of activities is packaged differently across lineages. *E. coli* keeps GTP cyclohydrolase II (*ribA*) and DHBP synthase (*ribB*) as **separate genes**, whereas *B. subtilis* fuses them into a **bifunctional RibA/RibBA** protein (PMID: 34569786, PMID: 8159171). This fusion/fission plasticity, plus the *ribG/ribD* and *ribC/ribF* naming divergence, is the most common source of cross-organism confusion.

Quaternary-structure divergence of lumazine synthase

Lumazine synthase is a textbook case of quaternary-structure evolution from an essentially fixed fold. **The protein exists as pentameric and icosahedral forms, built from practically the same structural monomeric unit** (PMID: 14660615), and across the Rhizobiales it is found **as free pentamers, decamers (dimers of pentamers), and icosahedrally arranged dodecamers of pentamers** (PMID: 17854827). The active sites lie at interfaces between adjacent subunits within a pentamer, so catalysis is conserved even as the higher-order assembly diverges. Some Rhizobiales encode two paralogs — an active Type I (RibH1) and a nearly inactive Type II (RibH2) — illustrating paralog subfunctionalization.

Modular supply: biosynthesis, uptake, and paralogs

Bacterial riboflavin supply is **modular**: cells obtain riboflavin **either through the endogenous riboflavin biosynthetic pathway (RBP) or via importer proteins, and frequently encode multiple paralogs of the RBP enzymes** (PMID: 27822970). Distinct *rib*-gene subsets appear to serve distinct functions — intracellular cofactor supply versus production of secreted flavins used, for example, in extracellular electron transfer and iron acquisition (the iron-riboflavin metabolic axis, PMID: 30026736).

Physiological-state variation

Flux through the pathway is modulated by physiological context. In industrial *E. coli*, a single **His115Leu mutation in the bifunctional RibF** reduces kinase activity to ~55% and, together with up-regulation of the whole *rib* branch, causes riboflavin to accumulate — a natural example of how a point change in the activation enzyme reshapes flux (PMID: 25926527). Anaerobes such as Clostridia (PMID: 34245087) and the broader iron–riboflavin axis further tie flavin output to redox and iron-limitation states.

6. Constraints, Dependencies, and Failure Modes

Ordering constraints (what must happen in what order)

- **RibA is obligatorily first.** Nothing downstream can proceed until GTP is committed and the pyrimidine skeleton is formed. Formate and pyrophosphate release are intrinsic to this step (PMID: 34569786).
- **RibD deamination precedes reduction in bacteria.** This order is not arbitrary — it is the diagnostic difference from fungi (PMID: 15208317). A path that reduced first would be the fungal, not the bacterial, route.
- **Both branches must feed RibH before RibE can act.** The pyrimidine (from RibA→RibD) and DHBP (from RibB) must both be present for lumazine synthesis; RibE then requires *two* lumazines per turnover.
- **Activation is strictly sequential:** riboflavin → FMN → FAD. FMN is the obligate intermediate; FAD cannot form without prior phosphorylation.

Metal and cofactor dependencies

- RibA requires **both Zn and Mg**; loss of either metal function would stall the entry step (PMID: 34569786).
- RibB requires **Mg** (PMID: 35802469).
- RibF/FADS requires **ATP and Mg** at both modules.

Regulatory failure modes and feedback

Feedback is exerted mainly at the RNA level. **FMN-specific riboswitches (RFN elements) direct expression of bacterial genes involved in the biosynthesis and transport of riboflavin** (PMID: 19169240), binding FMN directly through a six-stem junction with Mg²⁺-mediated phosphate contacts. Unusually, **the *E. coli* ribB FMN riboswitch operates at both the transcriptional and the translational level** (PMID: 25661987), giving dual-layer control over the carbohydrate-branch gene.

Layered on top of the riboswitch are protein-level inputs. In *B. subtilis*, the minor cryptic flavokinase **RibR specifically interacts *in vivo* with the non-translated wild-type leader of the rib mRNA** (PMID: 17590224), and genetic evidence indicates **additional FMN-dependent regulation beyond the core riboswitch** (PMID: 18666549). Regulatory loci such as *ribU/ribC* further modulate flavinogenesis, and the flavokinase/FAD-synthase gene itself feeds back on riboflavin accumulation (PMID: 19947554).

Because riboswitches are the control node, they are also a **failure/attack point**: the natural riboflavin analog **roseoflavin** is converted to roseoflavin mononucleotide, binds FMN riboswitch aptamers, and represses the pathway/transport genes (PMID: 19246992, PMID: 21593602). Resistance arises through aptamer mutations, and some organisms (*Streptomyces davawensis*) evolved a specialized riboswitch that discriminates FMN from roseoflavin mononucleotide (PMID: 22740651).

Selective druggability as an "engineered" failure mode

Because the pathway is absent in humans and animals, and because **many pathogens (including Gram-negative bacteria and yeast) lack an efficient riboflavin-uptake system and depend on endogenous biosynthesis, inhibiting these enzymes gives selective toxicity to the pathogen and not the host** (PMID: 30927319). This is validated experimentally: ribityllumazine/purinetrione analogues with alkyl-phosphate substituents **inhibit both *B. subtilis* lumazine synthase and *E. coli* riboflavin synthase**, and phosphonate versions reach **low-nanomolar Ki against *M. tuberculosis* lumazine synthase** (PMID: 16277343). Pathogen lumazine synthases have been characterized structurally for design, including *Candida albicans* LS (PMID: 17446177) and docking-guided imidazopyridines against *M. tuberculosis* LS (PMID: 25400412).

7. Controversies and Open Questions

Strongly supported claims. The reaction order, the identities and activities of each enzyme, the tandem-metal mechanism of RibA, the Mg-dependent formate-excision chemistry of RibB, the dismutation mechanism of RibE, the nested LS/RS capsid architecture, the bifunctional split-ancestry FADS, and FMN-riboswitch control are all backed by crystallography, enzymology, and genetics across multiple organisms.

Where the literature is thinner or organism-specific.

- **Substrate channeling in the LS/RS capsid.** The elegant geometry — a 60-mer RibH shell enclosing a RibE trimer — is well documented in *B. subtilis* (PMID: 7473709), but not all bacteria assemble this complex (many RibH forms are pentamers/decamers that do not encapsulate RibE). Whether channeling is functionally required, or merely one solution among several, remains open.
- **The physiological meaning of LS oligomeric diversity.** Why the same monomer builds pentamers, decamers, and icosahedra — and whether higher-order assembly confers stability, regulation, or moonlighting roles — is not resolved (PMID: 14523158, PMID: 29763607).
- **FADS module cross-talk.** The "dimer-of-trimers" model in which one protomer's FMNAT site is influenced by a neighbor's RFK site is derived largely from one organism (*C. ammoniagenes*) and partly from docking/MD rather than a co-crystal with substrates (PMID: 30177440, PMID: 25801930). Its generality across bacteria is untested.
- **Accessory regulation.** The precise mechanism by which RibR and *ribU/ribC*-class factors act, and how they integrate with the riboswitch, is described phenomenologically rather than structurally (PMID: 17590224, PMID: 19947554).
- **Scope limitation — Archaea.** The findings assembled here are drawn overwhelmingly from bacteria (and, for reaction order, plants and fungi). Archaeal riboflavin biosynthesis uses variant enzymes (e.g., distinct reductase/deaminase arrangements and archaeal-specific riboflavin kinases) and should not be assumed to follow the bacterial scheme.

Most important open questions. 1. Is the encapsulated LS/RS complex functionally channeled, and does disrupting encapsulation impair flux *in vivo*? 2. What determines LS quaternary state, and can it be engineered without altering catalysis? 3. Is FADS inter-module cross-talk conserved beyond corynebacteria, and can the non-homologous FMNAT module be selectively drugged in diverse pathogens? 4. How do protein-level regulators (RibR) and riboswitches co-regulate flux quantitatively under changing FMN demand?

8. Limitations and Knowledge Gaps

- **Organism sampling bias.** Mechanistic detail is concentrated in a handful of models (*E. coli*, *B. subtilis*, *C. ammoniagenes*, *M. tuberculosis*, Rhizobiales). Extrapolation to all bacteria — and especially to Archaea — is not warranted.
- **Nomenclature hazards.** Gene-name divergence (*ribD/ribG*, *ribF/ribC*, RibA vs. RibA/RibBA) means cross-study synthesis risks mis-assigning activities; this review maps by activity rather than gene letter.
- **Structural gaps.** No substrate-bound co-crystal of a full-length bacterial FADS is available; the FMNAT catalytic model rests partly on docking/MD.
- **In vivo vs. in vitro.** Much enzymology (K_i values, oligomeric equilibria) is from purified proteins; the *in vivo* relevance of transient FADS oligomers and of LS/RS encapsulation is inferred rather than directly demonstrated.
- **This was a literature-synthesis investigation**, not a primary-data analysis; conclusions are only as strong as the cited primary and review literature.

9. Proposed Follow-up Experiments / Actions

1. **Test channeling directly.** Engineer *B. subtilis* (or a tractable pathogen) RibH variants that retain catalysis but cannot encapsulate RibE, and measure riboflavin flux and lumazine leakage *in vivo* to determine whether encapsulation is functionally required.
2. **Cross-organism FADS survey.** Solve or model substrate-bound structures of full-length FADS from phylogenetically diverse pathogens to test whether the "dimer-of-trimers" cross-talk is general, and to guide selective FMNAT inhibitor design (PMID: 30177440).
3. **Archaeal scope study.** Characterize the reductase/deaminase order and kinase enzymology in representative Archaea to define where the bacterial model breaks down.
4. **Dual riboswitch + enzyme targeting.** Combine roseoflavin-class riboswitch ligands with nanomolar LS/RS inhibitors (PMID: 16277343) to probe synergy and resistance, exploiting that pathogens lacking uptake cannot bypass biosynthesis (PMID: 30927319).
5. **Quantitative regulatory model.** Reconstruct the FMN-riboswitch + RibR + FADS-feedback network to predict flux under varying flavoprotein demand, integrating the dual-level *ribB* riboswitch behavior (PMID: 25661987).

10. Key References

- **RibA mechanism / tandem metals & B. subtilis architecture:** [PMID: 34569786](#); [PMID: 8159171](#)
- **RibD bifunctional deaminase/reductase (drug target):** [PMID: 23722836](#)
- **RibB DHBP synthase chemistry:** [PMID: 35802469](#)
- **LS/RS ~1 MDa nested complex:** [PMID: 7473709](#)
- **Riboflavin synthase dismutation & lumazine synthase mechanism:** [PMID: 7893702](#)
- **Bifunctional FADS / split ancestry:** [PMID: 22892871](#); [PMID: 26627660](#); [PMID: 32466340](#); [PMID: 30177440](#); [PMID: 28341845](#); [PMID: 25801930](#); [PMID: 28779158](#)
- **FMN riboswitch regulation:** [PMID: 19169240](#); [PMID: 25661987](#); [PMID: 19246992](#); [PMID: 22740651](#); [PMID: 21593602](#)
- **Accessory regulation (RibR, ribU/ribC):** [PMID: 17590224](#); [PMID: 18666549](#); [PMID: 19947554](#)
- **Evolution / reaction order (bacteria vs. plant vs. fungi):** [PMID: 15208317](#)
- **LS quaternary-structure divergence:** [PMID: 14660615](#); [PMID: 17854827](#); [PMID: 14523158](#); [PMID: 29763607](#)
- **Modular supply / uptake / paralogs & iron axis:** [PMID: 27822970](#); [PMID: 30026736](#)
- **Druggability / inhibitors:** [PMID: 30927319](#); [PMID: 16277343](#); [PMID: 17446177](#); [PMID: 25400412](#)
- **Flux / engineering context:** [PMID: 25926527](#); [PMID: 25023436](#)
- **Adjacent / separable systems (do not conflate):** [PMID: 32570108](#) (F420); [PMID: 40044155](#) (triazines)

Prepared as a commissioned review synthesis. Claims are attributed to the primary and review literature cited above; uncertainty and organism-specific limits are flagged in Sections 7–8. The synthesis is bacteria-centric by design and should not be extrapolated to Archaea without direct evidence.